

Assignment: The Architect's Multiplication Engine (V2)

Python Programming - Week 8

1. The Story

In the Great Library of Algorithmia, the Architects have upgraded their manual "Shift-and-Add" engine to a new **Divide and Conquer** model. This model is faster but requires the engine to display its "recursive work" to ensure the sub-calculations are correct.

2. Problem Statement

Given two non-negative integers M and N , implement the **Karatsuba Algorithm**. To demonstrate the "intermediate steps" of this optimized approach, your program must output the three primary sub-products generated at the **top level** of the recursion.

- Let M and N be split into high and low parts: $M = m_1 \cdot 10^k + m_0$ and $N = n_1 \cdot 10^k + n_0$.
- You must output the results of:
 1. $z_0 = m_0 \cdot n_0$
 2. $z_1 = (m_0 + m_1) \cdot (n_0 + n_1)$
 3. $z_2 = m_1 \cdot n_1$
- **Efficiency:** Your implementation must aim for a time complexity better than $O(d_1 \cdot d_2)$.

3. Technical Requirements

- Output z_0, z_1, z_2 for the initial call (the first split of the input numbers).
- Output the final product $M \times N$.
- Include a comment block explaining the recurrence relation $T(n) = 3T(n/2) + O(n)$.

4. Example

Input:

1234

5678

Output (Intermediate values for top-level split):

z0: 23052 (34×78)

z1: 5976 ($((12+34) \times (56+78))$)

z2: 672 (12×56)

Result: 7006652

5. Implementation Skeleton

Submit as `week8_[yourRollnumber].py`.

```
1 """
2 Time Complexity Analysis:
3 Explain why Karatsuba is  $O(n^{1.58})$  and how it differs from
4 the  $O(n^2)$  digit-wise multiplication.
5 """
6
7 def karatsuba(m, n):
8     # Base case for small numbers
9     if m < 10 or n < 10:
10         return m * n
11
12     # Calculate the size of the numbers
13     n_digits = max(len(str(m)), len(str(n)))
14     half = n_digits // 2
15
16     # Split the numbers
17     # TODO: Implement split and recursive calls for z0, z1, z2
18
19     # Only print z0, z1, z2 for the FIRST call (depth 0)
20     # to satisfy the "Intermediate Values" requirement.
21
22     return final_result
23
24 if __name__ == "__main__":
25     m = int(input().strip())
26     n = int(input().strip())
27     print(karatsuba(m, n))
```